COS30019 – Final Assessment

Answering Instructions:

Please do not use a red pen/type in red.

There are 6 problems.

Total marks on paper: 118

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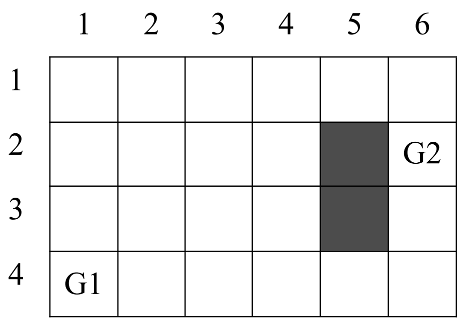
Problem 1 – Multiple-choice questions (15 marks)
(You can choose more than one option.)

1. Consider the maze below for the robot navigation problem, where being at either G1 or G2 satisfies the goal test. Shaded cells represent the wall. The robot can only travel north, south, east or west.



Assume that MD(*S1*, *S2*) is the Manhattan Distance between two squares *S1* and *S2*, and max[*x*, *y*] returns the biggest value of *x* and *y*, and min[*x*, *y*] returns the smallest value of *x* and *y*. Which of the following heuristic function for a square *S* is admissible?

A. h(*S*) = MD(*S*, G1)
B. h(*S*) = min[MD(*S*, G1), MD(*S*, G2)]
C. h(*S*) = max[MD(*S*, G1), MD(*S*, G2)]
D. h(*S*) = MD(*S*, G1) + MD(*S*, G2)

Answer: B

(5 marks)

2. Consider the Vacuum Cleaner World discussed in the lecture. If the agent does not know the geography of the environment and a room may become dirty again after being cleaned then what type of agent would best be used?

A. Simple reflex agent
B. Utility-based agent with learning
C. Goal-based agent
D. Rule-based agent with internal states.

Answer: A

(5 marks)

3. Which of the following statements can be used to correctly describe various paradigms of Artificial Intelligence (AI)?

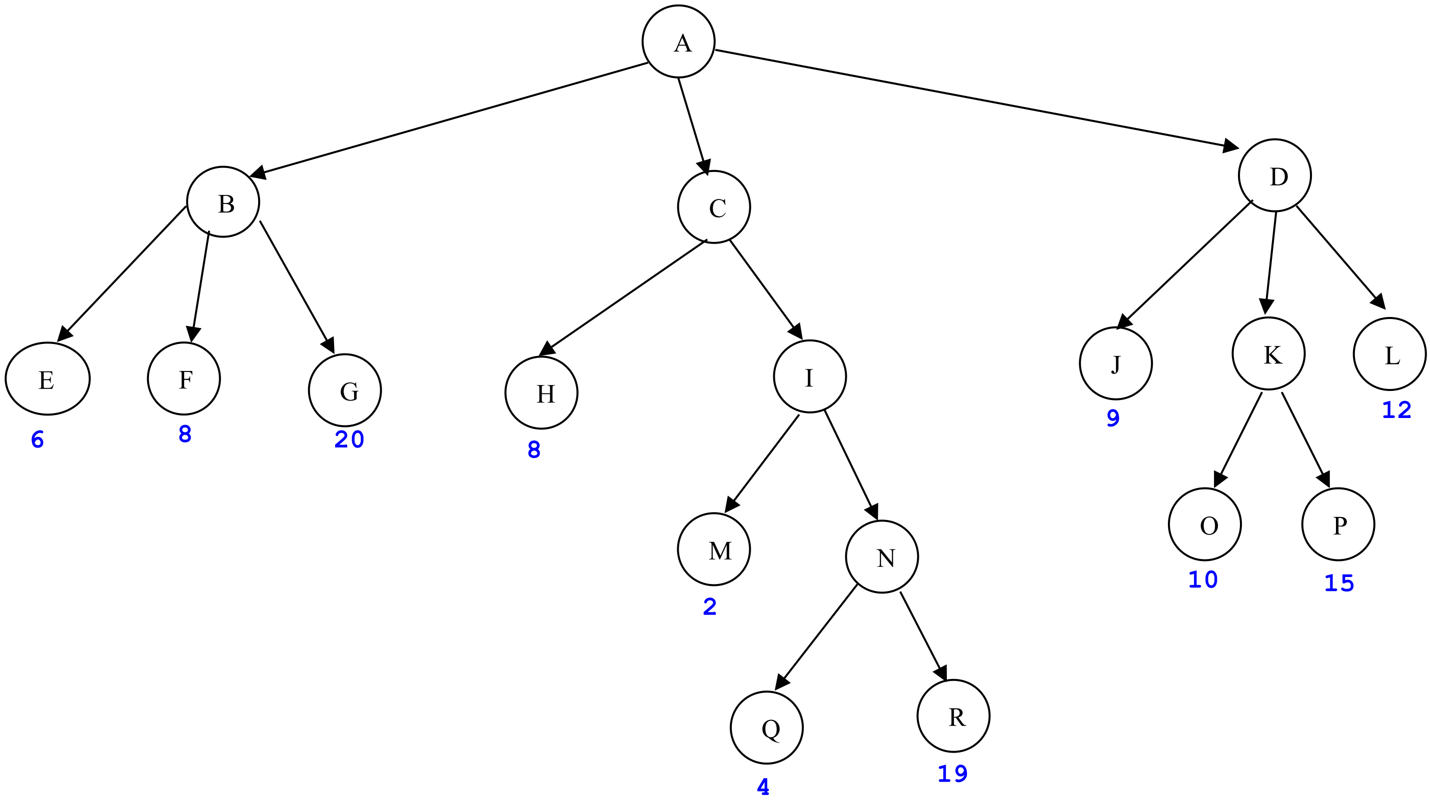
A. AI as a system that acts rationally aims to pass the Turing test.
B. AI as a system that thinks rationally aims to maximise the value of the performance measure.
C. AI as a system that thinks like a human always follows logical reasoning.
D. AI as a system that acts like a human aims to create machines that perform functions that require intelligence when performed by people.

Answer: D

(5 marks)

Problem 2 – Games and Adversary Search (23 marks)

Consider the following game tree in which the utility values are shown below each leaf node.



Assume that the root node corresponds to the maximising player. That is, **the first player (MAX)** is trying to maximise the final score. Assume that **the search always visits children left-to-right**.

State whether the following statements are TRUE or FALSE:

1. Node **B** is a MIN node: TRUE
2. Node **D** is a MAX node: FALSE
3. Node **I** is a MIN node: FALSE
4. Node **K** is a MAX node: TRUE
5. Node **N** is a MIN node: TRUE
6. According to the MINIMAX algorithm, the value of node **N** is 19: FALSE
7. According to the MINIMAX algorithm, the value of node **I** is 2: FALSE
8. According to the MINIMAX algorithm, the value of node **B** is 6: TRUE
9. According to the MINIMAX algorithm, the value of node **C** is 8: FALSE

10. According to the MINIMAX algorithm, the value of node **K** is 15: TRUE
11. According to the MINIMAX algorithm, the value of node **D** is 12: TRUE
12. According to the MINIMAX algorithm, the value of node **A** is 20: FALSE
13. According to the Alpha-Beta pruning algorithm, node **G** is pruned: FALSE
14. According to the Alpha-Beta pruning algorithm, node **C** is pruned: TRUE
15. According to the Alpha-Beta pruning algorithm, node **I** is pruned: FALSE
16. According to the Alpha-Beta pruning algorithm, node **N** is pruned: FALSE
17. According to the Alpha-Beta pruning algorithm, node **Q** is pruned: FALSE
18. According to the Alpha-Beta pruning algorithm, node **R** is pruned: TRUE
19. According to the Alpha-Beta pruning algorithm, node **D** is pruned: FALSE
20. According to the Alpha-Beta pruning algorithm, node **J** is pruned: FALSE
21. According to the Alpha-Beta pruning algorithm, node **K** is pruned: FALSE
22. According to the Alpha-Beta pruning algorithm, node **L** is pruned: FALSE
23. According to the Alpha-Beta pruning algorithm, node **P** is pruned: TRUE

(23 marks)

Problem 3 – Propositional Logic (24 marks)

$\neg \vee \wedge \neg \Rightarrow$

About my friend Bob, we know the following (the agent's knowledge base **KB**):
If Bob is having high blood pressure and Bob is having high cholesterol, then Bob is having a coronary artery disease (CAD).
Bob cannot be both having a CAD and not having a heart disease.
Bob is not having a heart disease.

We will use the following vocabulary:
HBP = "*Bob is having high blood pressure*"
HC = "*Bob is having high cholesterol*"
CAD = "*Bob is having a CAD*"
HD = "*Bob is having a heart disease*"

Part A: Determine which of the following propositional logic sentences correctly represent a sentence in the above knowledge base **KB**:

1. $(HBP \lor HC) \Rightarrow CAD$
2. $(HBP \land HC) \Rightarrow CAD$
3. $\neg(CAD \land \neg HD)$
4. $\neg CAD \land \neg HD$
5. $\neg HD$
6. $\neg CAD \lor HD$
7. $\neg HBP \lor \neg HC \lor CAD$
8. $\neg HBP \land \neg HC \lor CAD$
9. $\neg(HBP \land HC) \lor CAD$

(9 marks)

Hint: You may have to use logical equivalence for some questions.

Your answers to the above questions: My choices are 2,3,6,7,9 and 5

Part B: Using a truth table, state whether the following statements are TRUE or FALSE (the symbol $\models$ means **entailment**):

1. **KB** has exactly one model **[HBP=true, HC=true, CAD=true, HD=true]:** FALSE
2. **KB** has exactly one model **[HBP=false, HC= false, CAD= false, HD= false]**: FALSE
3. **KB** has exactly one model **[HBP=true, HC=true, CAD=true, HD= false]:** TRUE
4. **KB** has exactly two models **[HBP=true, HC=true, CAD=true, HD= false]** and **[HBP=false, HC= false, CAD= false, HD= false]:** FALSE
5. **KB** has exactly two models **[HBP=false, HC=true, CAD=false, HD= false]** and **[HBP=false, HC= false, CAD= false, HD= false:** FALSE
6. **KB** has exactly two models **[HBP=false, HC=true, CAD=false, HD= false]** and **[HBP=false, HC= false, CAD= false, HD= false]:** FALSE
7. **KB** has exactly three models **[HBP=false, HC=true, CAD=false, HD= false], [HBP=true, HC= false, CAD= false, HD= false]** and **[HBP=false, HC= false, CAD= false, HD= false]:** TRUE
8. **KB** has exactly three models **[HBP=true, HC=true, CAD=false, HD= false], [HBP=true, HC= false, CAD= false, HD= false]** and **[HBP=false, HC= false, CAD= false, HD= false]:** FALSE
9. **KB** has exactly four models **[HBP=false, HC=true, CAD=false, HD= false], [HBP=true, HC= false, CAD= false, HD= false], [HBP=true, HC= true, CAD= false, HD= false],** and **[HBP=false, HC= false, CAD= false, HD= false] :** FALSE
10. **KB** $\models \neg CAD$: FALSE
11. **KB** $\models$ HBP: FALSE
12. **KB** $\models$ HC: FALSE

13. KB $\models \neg HBP$: FALSE

14. KB $\models \neg HC$: FALSE

15. KB $\models \neg HBP \vee \neg HC$: TRUE

(15 marks)

Problem 4 – Propositional Logic (4x4 = 16 marks)

Using a truth table, please answer the following multiple-choice questions:

1. $(\neg A \vee B) \wedge ((\neg B \vee C) \wedge (A \wedge \neg C))$

(4 marks)

   a. The above sentence is unsatisfiable and has 0 model.
   b. The above sentence is valid and has 8 model.
   c. The above sentence is satisfiable and has 4 model.
   d. The above sentence is satisfiable and has 1 model.

ANSWER: a

2. $(A \wedge B) \vee (\neg A \vee \neg B)$

(4 marks)

   a. The above sentence is unsatisfiable and has 0 model.
   b. The above sentence is valid and has 4 model.
   c. The above sentence is satisfiable and has 2 model.
   d. The above sentence is satisfiable and has 1 model.

ANSWER: b

3. $((A \Rightarrow B) \wedge (B \Rightarrow C)) \wedge (A \vee \neg C)$

(4 marks)

   a. The above sentence is unsatisfiable and has 0 model.
   b. The above sentence is valid and has 8 model.
   c. The above sentence is satisfiable and has 1 model.
   d. The above sentence is satisfiable and has 2 model.
   e. The above sentence is satisfiable and has 4 model.

ANSWER: d

4. $((A \Leftrightarrow B) \wedge (B \Leftrightarrow C)) \wedge ((A \Leftrightarrow C) \wedge A)$

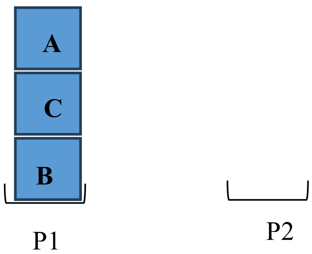
(4 marks)

   a. The above sentence is unsatisfiable and has 0 model.
   b. The above sentence is valid and has 8 model.
   c. The above sentence is satisfiable and has 1 model.
   d. The above sentence is satisfiable and has 2 model.
   e. The above sentence is satisfiable and has 4 model.

ANSWER: c

Problem 5 – AI Planning (20 marks)

Our agent is a robot with two hands: *Hand1* and *Hand2*. The robot's task is to stack the blocks labelled by the letters A, B, C to achieve a desired configuration. There are two places, *P1* and *P2*, where a block can occupy, and a block can also be put on top of another block. Once a place is occupied by a block, no other thing can be put there. Initially, block B occupies place P1 while place P2 is unoccupied, and block C is on block B, and block A is on block C (see below figure).



The goal is to achieve a configuration in which block C occupies place P1, and block B is on block C, and block A is on block B.

A block is clear if there is nothing on top of it. (For instance, in the above figure, only block A is clear).

The actions available to the robot are *PickUp* and *PutDown. PickUp* a clear block results in holding that block using a free hand. One of the effects of *PickUp* is that the free hand that the robot uses to pick up the block will no longer be free after picking up the block. *PutDown* requires the robot to hold a block and the robot can only put it down either at an unoccupied place or on a clear block. Of course, after putting down a block, the robot's hand that held it would become free.

Hint: When answering the following question, you may need to represent an unoccupied place as being clear so that a block can be stacked on another block or on a place.

1. Complete the description of the initial state and the agent's goals.

(10 marks)

ANSWER: In my opinion, I would like to say:
- Initial State:
 Block Positions: On (B, P1) means Block B is above place P1
 On (C, B) means Block C is above Block B
 On (A,C) means Block A is above Block C
 Status: [P2, Clear] means Place2 is currently clear
 [A, Clear] means nothing is above on Block A
 Hand: [Hand1, Free] means Hand1 is not hold any blocks
 [Hand2, Free] means Hand1 is not hold any blocks
- Goal State: Block Position: On (C, P1) means Block C is above place P1
 On (B, C) means Block B is above Block C
 On (A, B) means Block A is above Block B
 Status: [P2, Clear] means Place2 is currently clear
 [A, Clear] means nothing is above on Block A.

2. Write down STRIPS-style definitions of the two actions.

(10 marks)

ANSWER: From my perspectives, the STRIPS-style of this actions:

1. [P2, A] means currently P2 contains Block A
2. Next, Command PICKUP with [Hand1, C] means uses hand 1 to pick up Block C
3. Then, Command PICKUP with [hand2, B] means uses hand2 to pick up Block B
4. Subsequently, Command PUTDOWN with [Hand1,C] in Status [C,P1] means that use hand1 to put down block C on place P1.
5. Command PUTDOWN [hand2,B] with status [B,C] means procees comman use hand2 to put down Block B on Block C
6. Finally, status [A,B] means put Block A on Block B. Subsequently, the agent reach the goal status: [C,P1], [B,C], [C,A] and [A,Clear]

Problem 6 – Uncertain reasoning (20 marks)

A company launched and operates its satellite, KoalaSat, in space. KoalaSat is displaying some abnormal behaviour and the company's engineers speculate that it is because of a faulty power system. They run a diagnostic test on the power system, PSDT, to determine whether the satellite power system is faulty. The test PSDT would return either a **RedFlag** (Test determines Fault) or **GreenTick** (test finds No Fault). The accuracy of the PSDT test is as follows:

1. Given the satellite power system being faulty, the probability of the test returning **RedFlag** is 0.96;
2. Given the satellite power system being NOT faulty, the probability of the test returning **GreenTick** is 0.98.

After running the test on KoalaSat, the test returns **GreenTick**. According to the manufacturer of KoalaSat, at KoalaSat's current age, 1 in 100 satellites would have a faulty power system.

We will use the following vocabulary in the rest of this Problem:
Fault – The satellite KoalaSat's power system is faulty;
TestRF – The PSDT test on KoalaSat returns a **RedFlag**.

1. Which of the following equations can be used to represent the sentence: "*Given the satellite power system being faulty, the probability of the test returning **RedFlag** is 0.96*"?

   (5 marks)

   a. P(**Fault** ∧ **TestRF**) = 0.96
   b. P(**Fault** ⇒ **TestRF**) = 0.96
   c. P(**Fault** | **TestRF**) = 0.96
   d. P(**TestRF** | **Fault**) = 0.96
   e. P(**TestRF** = **Fault**) = 0.96

   ANSWER: d

2. After running the test on KoalaSat and it returns **GreenTick**, which of the following expression can be used to compute *the probability of KoalaSat's power system being faulty*?

   (5 marks)

   a. P(**Fault** | **TestRF**)
   b. P(**TestRF** | **Fault**)
   c. P(**Fault** | **1 - TestRF**)
   d. P(¬**TestRF** | **Fault**)
   e. P(**Fault** | ¬**TestRF**)

   ANSWER: e

3. Use Bayes' rule to calculate the probability of KoalaSat's power system being faulty. How do you interpret your results in English?

   Hint: For this question, you'll need to calculate the probability of KoalaSat's power system being faulty (using Bayes' rule and show your working). *Then, you'll need to write 1 to 2 sentences to interpret your results in English.*

(10 marks)

ANSWER: